LESSON PLAN · SCRATCH (GR 3–5) INTERMEDIATE: STORY, QUIZ & MUSIC STUDIO

Lesson 11: Keyboard Instruments

Module 4 | 45 minutes | Events and parallel scripts | Grades 3–5 · ages 8–10

Objective: Turn keys and sprites into a playable instrument with events, and sync animations to the music with messages.

Success criteria

- I can make keys play notes.
- I can make a sprite into an instrument.
- I can sync animation to a beat with messages.

Key vocabulary

- **instrument**: Something that makes musical sounds.
- **tempo**: How fast the beat is.
- **parallel**: Running at the same time.
- **sync**: Happening together in time.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (45 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Conductor game (5 min): one learner conducts; others clap only when the conductor's hand comes down (the message).
5 min	**Watch**	Play the lesson video with captions; pause at each heading and ask a question.
8 min	**Build along**	Build a keyboard piano: learners follow the steps.
14 min	**Perform**	Code Lab: keyboard piano
9 min	**Probe**	Code Lab: dance to the beat, Keep in time, activity
4 min	**Wrap-up**	Exit ticket: Write the script for the f key to play note 65.

Unplugged warm-up

Conductor game (5 min): one learner conducts; others clap only when the conductor's hand comes down (the message).

Teaching the idea

Each key event starts its own script, so the keyboard becomes an instrument. A broadcast-driven beat loop synchronises animations to music, illustrating parallel scripts coordinated by messages.

Analogy: The band leader taps the beat; every musician (sprite) listens and plays along.

Live demo: Build a 3-key piano live; then add a dancer with its own wait loop and show it drifting, then sync with a beat broadcast.

Questions to ask

Ask	Listen for
Why does each key need its own script?	Each key is a different event.
How do we keep the dancer in time?	Receive the beat message.
How do you change the tempo?	Change the wait in the beat loop.

Differentiation

Support
Provide the a/s/d scripts; learners add f and g.

Stretch
Add a “record” feature idea: a list is needed (next course!) — or play a pre-programmed song on the space key.

Common misconceptions

Misconception	How to fix it
Holding a key plays forever	Each press triggers the script once.
All sprites need their own beat loop	One leader loop with a broadcast keeps everyone together.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
Write the script for the f key to play note 65.

Homework
Teach someone at home to play a short tune on your keyboard piano.

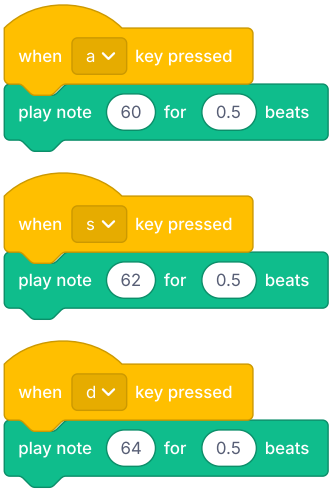
Answer key — Kitabash activities

1. Code Lab: keyboard piano (perform · code lab)

Goals checked: a plays a note · s plays a note · d plays a note

One solution:

Piano

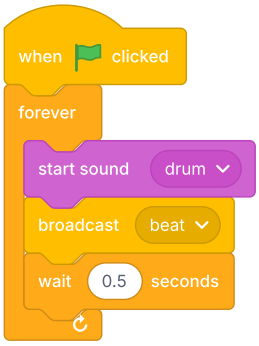


2. Code Lab: dance to the beat (probe · code lab)

Goals checked: Broadcasts “beat” · The Dancer dances in time · A drum keeps time

One solution:

Piano



Dancer



3. Keep in time (probe · multiple choice)

Answer: The Dancer always moves exactly on the beat

4. Explain your thinking (probe · free response)

Look for: key, note, click. **Example:** I would make a drum kit: the space key plays a big drum, the Ball sprite plays a clap when clicked, and keys a to g play a scale of notes for the tune.

Answer key — worksheet

1. **Key words:** instrument = Something that makes musical sounds. · tempo = How fast the beat is. · parallel = Running at the same time. · sync = Happening together in time.
2. **Plan your code:** Goals: a plays a note; s plays a note; d plays a note.
3. **Plan your code:** Goals: Broadcasts “beat”; The Dancer dances in time; A drum keeps time.
4. **Keep in time:** A. The Dancer always moves exactly on the beat — One loop keeps time; everything else follows it, so nothing drifts out of sync.
5. **Explain your thinking:** I would make a drum kit: the space key plays a big drum, the Ball sprite plays a clap when clicked, and keys a to g play a scale of notes for the tune.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.